

PS 3 Week 3 Section

Subsetting, Means, and One-/Two-Way Tables

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Week 3

Today's Agenda

- Vocabulary & key terms recap
- Functions
- Live demo: putting it together
- Common mistakes (and how to avoid them)
- Why do we care? Tables, hypotheses, and a caution

Goal: by the end of section, you can go from a raw dataset to an answer to a research question, using `$`, `mean()`, `subset()`, and `table()`.

Quick Vocabulary Check

- **Data frame:** a rectangular dataset in R – rows are observations, columns are variables (e.g. `cafe`).
- **Variable (column):** one measured characteristic, e.g. `drinks`.
- **Observation (row):** one unit in the dataset, e.g. one day/time slot.
- **Scalar:** a single value, e.g. 85, or the result of `mean(x)`.
- **Vector:** a sequence of values of the same type, e.g. `cafe$drinks`.
- **Categorical variable:** takes on named categories, e.g. "AM" / "PM".
- **Function / argument:** `mean(x)` is a function; `x` is the argument you give it.

Functions we have learned

Function	What it does	Example
<code>dataset\$var</code>	pulls out one column	<code>officerdata\$contra</code>
<code>mean(x)</code>	average of a numeric variable	<code>mean(officerdata\$contra)</code>
<code>subset(data, cond)</code>	keeps only rows meeting a condition	<code>subset(officerdata, officer_female == 1)</code>
<code>table(x)</code>	counts of one categorical variable	<code>table(officerdata\$driver_race)</code>
<code>table(x, y)</code>	counts of combinations of two categorical variables	<code>table(x, y)</code>

A New Example: Coffee Shop Data

Small dataset practice.

```
cafe <- data.frame(  
  day      = c("Mon", "Mon", "Tue", "Tue", "Wed", "Wed", "Fri", "Fri"),  
  daypart  = c("AM", "PM", "AM", "PM", "AM", "PM", "AM", "PM"),  
  drinks  = c(85, 40, 90, 45, 78, 50, 120, 80),  
  size_cat = c("Rush", "Slow", "Rush", "Slow", "Rush", "Slow", "Rush", "Slow")  
)  
head(cafe)
```

```
#>   day daypart drinks size_cat  
#> 1 Mon      AM     85     Rush  
#> 2 Mon      PM     40     Slow  
#> 3 Tue      AM     90     Rush  
#> 4 Tue      PM     45     Slow  
#> 5 Wed      AM     78     Rush  
#> 6 Wed      PM     50     Slow
```

Spot the error (or lack thereof)

Discuss these in small groups (2-4 people) and if you think it will run and if it does, what will it print out.

```
cafe.morning <- subset(cafe, daypart == "AM")
```

```
cafe$cafe.morning$drinks
```

```
mean(cafe)
```

Spot the error (or lack thereof)

```
cafe.morning <- subset(cafe, daypart == "AM")
```

Correct!

```
cafe$cafe.morning$drinks
```

Should just be `cafe.morning$drinks`

```
mean(cafe)
```

Needs to do a specific column in the dataset. Such as
`mean(cafe$drinks)`

Step 1: Pull Out a Variable and Take a Mean

```
cafe$drinks
```

```
#> [1] 85 40 90 45 78 50 120 80
```

```
mean(cafe$drinks)
```

```
#> [1] 73.5
```

What does this one number tell us and what does it hide?

Step 2: Subset, Then Summarize

```
morning.only <- subset(cafe, daypart == "AM")  
mean(morning.only$drinks)
```

```
#> [1] 93.25
```

- Same idea as Week 2: `subset()` first, *then* compute on the subset.
- What changes when we switch "AM" for "PM"?

What changes when we switch “AM” for “PM”?

```
afternoon.only <- subset(cafe, daypart == "PM")  
mean(afternoon.only$drinks)
```

```
#> [1] 53.75
```

Step 3: One-Way Table

```
table(caffe$size_cat)
```

```
#>
```

```
#> Rush Slow
```

```
#>    5    3
```

- Frequency table: how many rows fall into each category?
- Useful first step before you subset or cross-tabulate.

Step 4: Two-Way Table

```
table(cafes$daypart, cafes$size_cat)
```

```
#>
```

```
#>      Rush Slow
```

```
#> AM      4    0
```

```
#> PM      1    3
```

- Rows = first variable, columns = second variable.
- How many instances fall into combinations of categories?

Correcting Common Mistakes

```
# Mistake: single = instead of double ==  
subset(cafe, daypart = "AM")          # ERROR
```

```
# Fixed: use == to test equality, not =  
subset(cafe, daypart == "AM")        # correct
```

- **= assigns** a value; **== compares** two values.
- Only **==** belongs inside a `subset()` condition.

Mean of a Categorical Variable

```
mean(caffe$daypart)
```

```
#> Warning in mean.default(caffe$daypart): argument is not numeric
```

```
#> returning NA
```

```
#> [1] NA
```

- `daypart` is categorical ("AM" / "PM"), not numeric. So `mean()` gives a warning and returns NA
- Check whether a variable is categorical or numeric *before* you use `mean()`

Using the \$ to get a column from a dataframe

- Before the \$ is the name of the dataframe, after the \$ is the name of the column.
- If we wanted to get the days in the dataset, what would we type?
 - column names are: day, daypart, drinks, size_cat
 - name of the dataframe: cafe

Using the \$ to get a column from a dataframe

- If we wanted to get the days in the dataset, what would we type?

```
cafe$day
```

```
#> [1] "Mon" "Mon" "Tue" "Tue" "Wed" "Wed" "Fri" "Fri"
```

Why Do We Care? (Recap from Lecture)

Theory $\rightarrow$ Hypothesis $\rightarrow$ Empirical Test $\rightarrow$ Re-Evaluate Theory

- A one-way table describes *one* thing.
- A two-way table lets us **test a hypothesis** about how two variables relate.
- Example: “Rush periods are more common in the morning” is a hypothesis we can *check* with the two-way table above.

Tips for Analyzing the Data

- Stick to the facts first: what does the table literally say?
- *Then* you might hypothesize about *why*.
- **We only know what the data shows.** The data can't tell us why something is happening.
 - E.g., with the officer-stop data from Week 2: if female officers have a higher contraband-find rate than male officers, is that because they're better at spotting contraband, or because they're more selective about who they search? The table alone can't say.

Reminders

- Make sure to follow the instructions at the end of the assignments before you submit
 - If you forget to save and wait 5 seconds, you may lose points because it does not have the last things you typed in
- Make sure to use the same names of dataframes and variables as the original .ipynb file has – that is how the autograder is checking your work

Recap

Function	Use it when...
<code>dataset\$var</code>	you need one column
<code>mean()</code>	you want the average of a numeric variable
<code>subset()</code>	you want only some rows
<code>table(x)</code>	you want counts of one categorical variable
<code>table(x, y)</code>	you want to see how two categorical variables relate

Before You Go

- You have the remainder of class to work on the problem set.
- **Problem Set 1: due October 2.**
- R Cheat Sheet has every function from today.
- My office hours: **Mon & Fri, 1–2pm.**
- Check the syllabus for other GSIs' office hours.

Attendance

